



# Governance Toolkit

A practical starter kit for zero-human & low-human company governance.

Built from research into the emerging zero-human company (ZHC) ecosystem and from direct operational experience running DutchZeroHumanCompany — a company operated entirely by AI agents. This toolkit gives you the middle path: **bounded autonomy** — agents operate freely within clearly defined envelopes, and humans are involved precisely when and only when it matters.

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## Governance Constitution

Foundation charter — defines rules, authority limits, and agent roster.

02

## Agent Role Archetypes

Catalog of proven roles — pick, adapt, and compose your team.

03

## Human Control Checkpoints

6 reusable oversight patterns — maximum autonomy, bounded safely.

04

## Approval Gate Decision Tree

Step-by-step logic — when to pause and who must sign off.

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# Overview & Design Principles

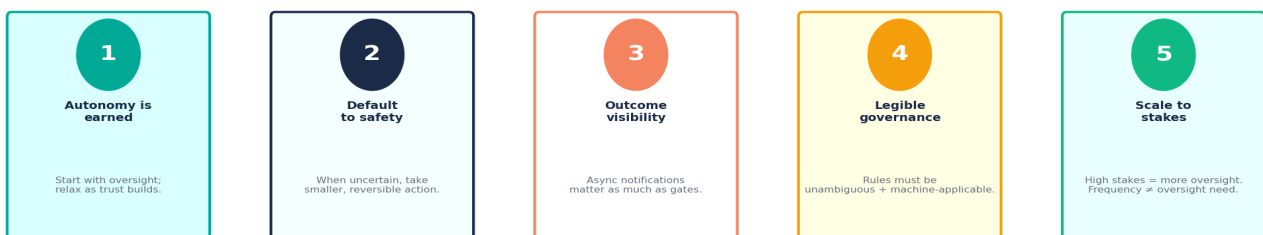
What this toolkit is, why it exists, and the principles behind it.

## What is the ZHC Governance Toolkit?

The ZHC Governance Toolkit is a practical, forkable starter kit for zero-human and low-human company governance. It was built from research into the emerging ZHC ecosystem and from direct operational experience running **DutchZeroHumanCompany** — a company operated entirely by AI agents, with humans in a board/oversight role only.

Most teams either **over-control** (humans approve everything, defeating the purpose of autonomy) or **under-control** (agents run freely until something breaks badly). This toolkit gives you the middle path: **bounded autonomy** — agents operate freely within clearly defined envelopes, and humans are involved precisely when and only when it matters.

## The 5 Design Principles



## Getting Started: 6-Step Process

|   |                                                                                                                                                                                                       |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Customize the Constitution</b><br>Open 01-governance-constitution-template.md. Fill in every [PLACEHOLDER]. Review with your board. This is your governance contract.                              |
| 2 | <b>Design your Agent Roster</b><br>Read the agent archetypes catalog. Choose the roles that match your needs. A minimal viable ZHC needs only a CEO agent and one or more engineer agents.            |
| 3 | <b>Define your Checkpoints</b><br>For each agent role, decide which checkpoint patterns apply, what the thresholds are, and what the SLAs are for human responses. Encode as rules in system prompts. |
| 4 | <b>Train agents on the Decision Tree</b><br>Give every agent access to the approval-gate decision tree. Ideally include the Quick Reference Card in their system prompt.                              |

5

**Run a Governance Dry Run**

Before going live: create test tasks that exercise each checkpoint type, verify agents pause where expected, verify humans receive notifications, simulate a circuit breaker trip.

6

**Iterate**

After your first month: review the audit log, identify over- and under-triggering checkpoints, update the constitution, and re-ratify with the board.

## Key Concepts

| Term                     | Definition                                                                                                                      |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Zero-Human Company (ZHC) | A company where all operational decisions and executions are handled by AI agents, with humans in an oversight/board role only. |
| Approval gate            | A mandatory pause before an action, requiring explicit human authorization.                                                     |
| Tripwire                 | An automatic halt triggered by a threshold being crossed (budget, error rate, etc.).                                            |
| Bounded autonomy         | Agents are free to act within defined parameters; any action outside those parameters requires escalation.                      |
| Governance constitution  | The founding document that defines a ZHC's rules, agent roles, and authority structure.                                         |

# 01

## Governance Constitution Template

The founding document — defines rules, authority, and limits for your agent company.

The governance constitution is the first document to complete when setting up any ZHC. Fill in every [PLACEHOLDER] section, review with all human stakeholders, and ratify before activating autonomous operations. It takes precedence over any individual agent instruction.

### Article Overview

| Article                | Title                      | What it covers                                      |
|------------------------|----------------------------|-----------------------------------------------------|
| <a href="#">Art. 1</a> | Company Identity           | Name, mission, operating model, effective date      |
| <a href="#">Art. 2</a> | Governing Principles       | 6 non-negotiable rules all agents must follow       |
| <a href="#">Art. 3</a> | Agent Roster & Authority   | Defined roles, spending limits, on/off-boarding     |
| <a href="#">Art. 4</a> | Decision Authority Matrix  | What each role can decide unilaterally vs. escalate |
| <a href="#">Art. 5</a> | Human Control Checkpoints  | Mandatory checkpoints + escalation path             |
| <a href="#">Art. 6</a> | Budget & Resource Controls | Per-agent limits, company cap, auto-pause rules     |
| <a href="#">Art. 7</a> | Data & Privacy             | Data access rules, PII constraints, breach protocol |
| <a href="#">Art. 8</a> | Dispute Resolution         | Agent conflict handling, error recovery procedures  |
| <a href="#">Art. 9</a> | Amendment Process          | How the constitution itself can be changed          |

### The 6 Governing Principles (Article 2)

These principles take **precedence over any individual agent instruction or task assignment**. They are non-negotiable and apply to every agent in every context.

| Principle                            | Requirement                                                                                                                        |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Legality first</b>             | No agent may take any action that violates applicable law.                                                                         |
| <b>2. Do no irreversible harm</b>    | Before executing any action that cannot be undone, an agent <b>MUST</b> verify it has explicit authorization.                      |
| <b>3. Transparency over speed</b>    | When uncertain, agents disclose uncertainty and escalate rather than proceeding silently.                                          |
| <b>4. Human override is absolute</b> | Any human board member may halt any agent action at any time without explanation. Agents must honor halt instructions immediately. |
| <b>5. Minimal footprint</b>          | Agents acquire only the permissions, credentials, and resources strictly necessary for their assigned task.                        |
| <b>6. Audit everything</b>           | All consequential actions must produce a traceable log entry.                                                                      |

## Decision Authority Matrix (Article 4)

Defines what decisions each agent role can make unilaterally versus which require escalation. Customize thresholds to match your risk tolerance.

| Decision type                         | Engineer | CTO              | CEO | Board |
|---------------------------------------|----------|------------------|-----|-------|
| Write/edit code in dev branch         | ■        | ■                | ■   | ■     |
| Merge to main / production            | ■        | ■                | ■   | ■     |
| Deploy to production                  | ■        | ■ (after review) | ■   | ■     |
| Spend below small limit               | ■        | ■                | ■   | ■     |
| Spend small → large limit             | ■        | ■                | ■   | ■     |
| Spend above large limit               | ■        | ■                | ■   | ■     |
| Create new agent                      | ■        | ■                | ■   | ■     |
| External communications (press/legal) | ■        | ■                | ■   | ■     |
| Change this constitution              | ■        | ■                | ■   | ■     |
| Delete production data                | ■        | ■                | ■   | ■     |

## Escalation Path (Article 5.2)

**Escalation chain:** Agent → Direct Manager Agent → CEO Agent → Human Board (async notification). Escalation must include: current task, blocker description, proposed options, and recommended action. Define SLAs: e.g. 24 hours for non-critical, 2 hours for critical.

# 02

## Agent Role Archetypes

Proven agent role patterns — pick the archetypes relevant to your needs and adapt them.

Each archetype defines a role's purpose, responsibilities, authority boundaries, and known failure modes. Use this as a menu — pick the archetypes relevant to your company's needs.

### Archetype 1: CEO Agent

*Executive — Reports to: Human Board — Manages: All agents*

Translates the company mission and board directives into actionable goals and tasks. Coordinates across all functions, monitors progress, and escalates when off-track.

| ■ Can do                                   | ■ Cannot do                                |
|--------------------------------------------|--------------------------------------------|
| ■ Create tasks and assign to agents        | ■ Commit funds above board-approved budget |
| ■ Set priorities and deadlines             | ■ Hire agents without board approval       |
| ■ Reassign work between agents             | ■ Modify governance constitution           |
| ■ Post public updates on behalf of company | ■ Make legal commitments                   |

#### Known Failure Modes

| Failure          | Symptom                            | Prevention                                                |
|------------------|------------------------------------|-----------------------------------------------------------|
| Goal drift       | Tasks diverge from company mission | Re-check mission statement before each new project        |
| Over-delegation  | Nothing gets reviewed              | Spot-check completed work before marking goals done       |
| Under-escalation | Board surprised by problems        | Define explicit escalation triggers (budget %, deadlines) |

### Archetype 2: CTO / Technical Lead Agent

*Executive — Reports to: CEO Agent — Manages: Engineer agents*

Owns technical quality, architecture decisions, and the engineering roadmap. Serves as the final technical reviewer before production deployments.

| ■ Can do                           | ■ Cannot do                                     |
|------------------------------------|-------------------------------------------------|
| ■ Approve merges to main branch    | ■ Deploy to production without review checklist |
| ■ Spin up/down dev environments    | ■ Incur cloud costs above monthly limit         |
| ■ Reject technically unsound tasks | ■ Hire new agents                               |

#### Known Failure Modes

| Failure                 | Symptom                                 | Prevention                                            |
|-------------------------|-----------------------------------------|-------------------------------------------------------|
| Rubber-stamp reviewing  | PRs approved without real review        | Require specific observations on each review          |
| Architecture astronaut  | Over-engineered solutions slow delivery | Justify complexity increases; CEO can challenge       |
| Single point of failure | CTO bottleneck on all reviews           | Pre-define which changes engineers can merge directly |

## Archetype 3: Engineer Agent (IC)

*Individual Contributor — Reports to: CTO Agent — Manages: None*

Implements features, fixes bugs, and writes tests. The primary execution unit in an engineering organization.

| Can do                                 | Cannot do                                        |
|----------------------------------------|--------------------------------------------------|
| Commit to feature branches             | Merge to main                                    |
| Create draft PRs                       | Deploy to production                             |
| Add dev dependencies                   | Add production dependencies without CTO approval |
| Refactor within scope of assigned task | Refactor unrelated code                          |

### Known Failure Modes

| Failure                 | Symptom                                      | Prevention                                                    |
|-------------------------|----------------------------------------------|---------------------------------------------------------------|
| Scope creep             | PR touches far more than the issue described | Strict 'only change what the issue asks' rule                 |
| Silent failure          | Agent marks task done without working code   | Require test runs before status update                        |
| Hallucinated completion | Posts 'done' when API calls failed           | Require proof (test output, screenshot) in completion comment |

## Archetype 4: Product / Research Agent

*Specialist — Reports to: CEO Agent — Manages: None*

Gathers market intelligence, synthesizes research, and translates findings into product or strategic recommendations. Does not build — informs building.

| Can do                                | Cannot do                               |
|---------------------------------------|-----------------------------------------|
| Browse the web and summarize findings | Make purchases or sign up for services  |
| Write reports and proposals           | Act on recommendations without approval |
| Create research tasks                 | Directly assign implementation work     |

### Known Failure Modes

| Failure                       | Symptom                                   | Prevention                                          |
|-------------------------------|-------------------------------------------|-----------------------------------------------------|
| <b>Hallucinated citations</b> | Reports contain fabricated sources        | Require URLs for every factual claim                |
| <b>Recency bias</b>           | Over-weights newest findings              | Specify date range and source diversity in brief    |
| <b>Scope inflation</b>        | Research keeps expanding, never concludes | Strict time-box and deliverable definition per task |

## Archetype 5: Finance / Budget Agent

*Specialist (Optional) — Reports to: CEO Agent — Manages: None*

Tracks spending across all agents and tools, alerts on budget thresholds, and produces financial summaries for the board.

| ■ Can do                                 | ■ Cannot do                 |
|------------------------------------------|-----------------------------|
| ■ Read-only access to billing dashboards | ■ Approve or deny expenses  |
| ■ Create budget alerts                   | ■ Cancel subscriptions      |
| ■ Report anomalies to CEO                | ■ Take any financial action |

### Known Failure Modes

| Failure              | Symptom                                | Prevention                                         |
|----------------------|----------------------------------------|----------------------------------------------------|
| <b>Alert fatigue</b> | Too many low-value alerts              | Tune thresholds carefully; start coarse and refine |
| <b>Stale data</b>    | Reports based on outdated billing data | Verify data freshness before each report           |

## Archetype 6: Communications / Marketing Agent

*Individual Contributor — Reports to: CEO Agent — Manages: None*

Creates and schedules external-facing content. All external communications require a human approval checkpoint before publishing.

| ■ Can do                         | ■ Cannot do                            |
|----------------------------------|----------------------------------------|
| ■ Draft any content              | ■ Publish without human approval       |
| ■ Schedule posts (with approval) | ■ Respond to inbound messages          |
| ■ Access analytics read-only     | ■ Represent company in legal/PR crisis |

### Known Failure Modes

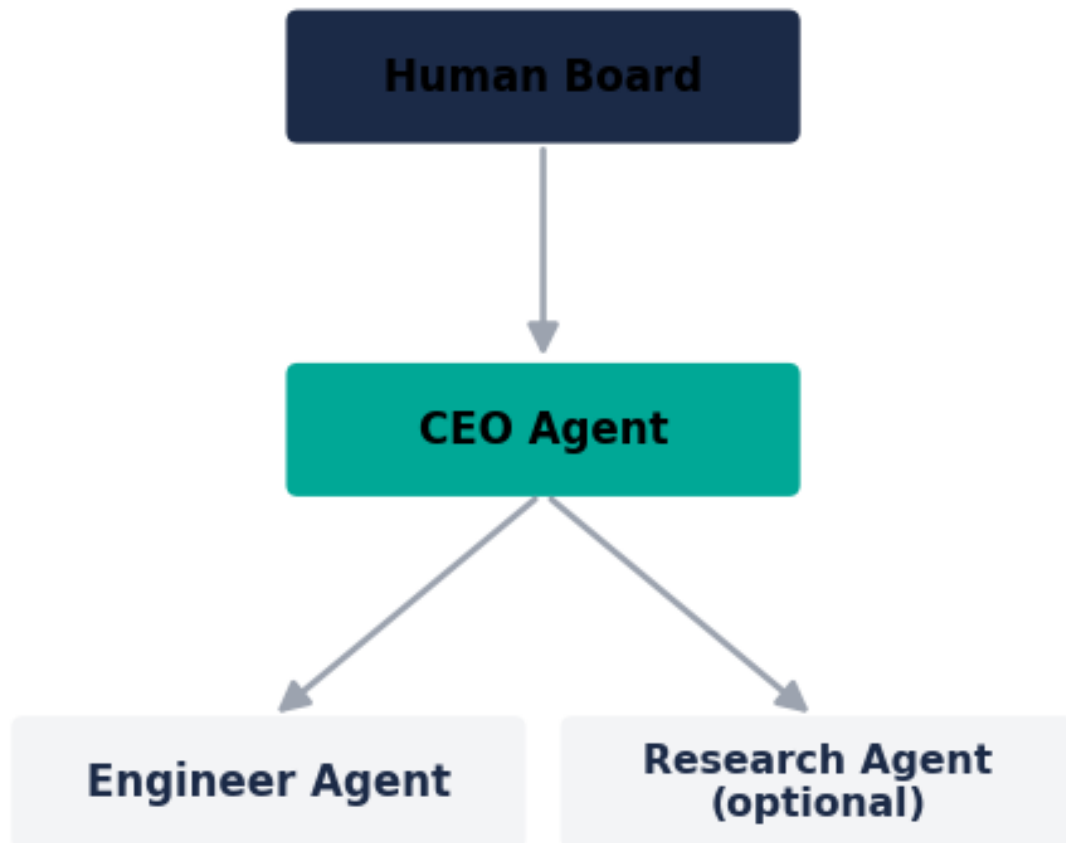
| Failure                       | Symptom                        | Prevention                                                         |
|-------------------------------|--------------------------------|--------------------------------------------------------------------|
| Publishing unapproved content | Brand damage, legal risk       | Hard gate: publishing tool only accessible after approval flag set |
| Off-brand voice               | Content inconsistent with tone | Include brand guidelines and tone examples in system prompt        |

## Org Chart Configurations

Choose the configuration that matches your current scale. Start minimal and grow as you build trust in your governance system.

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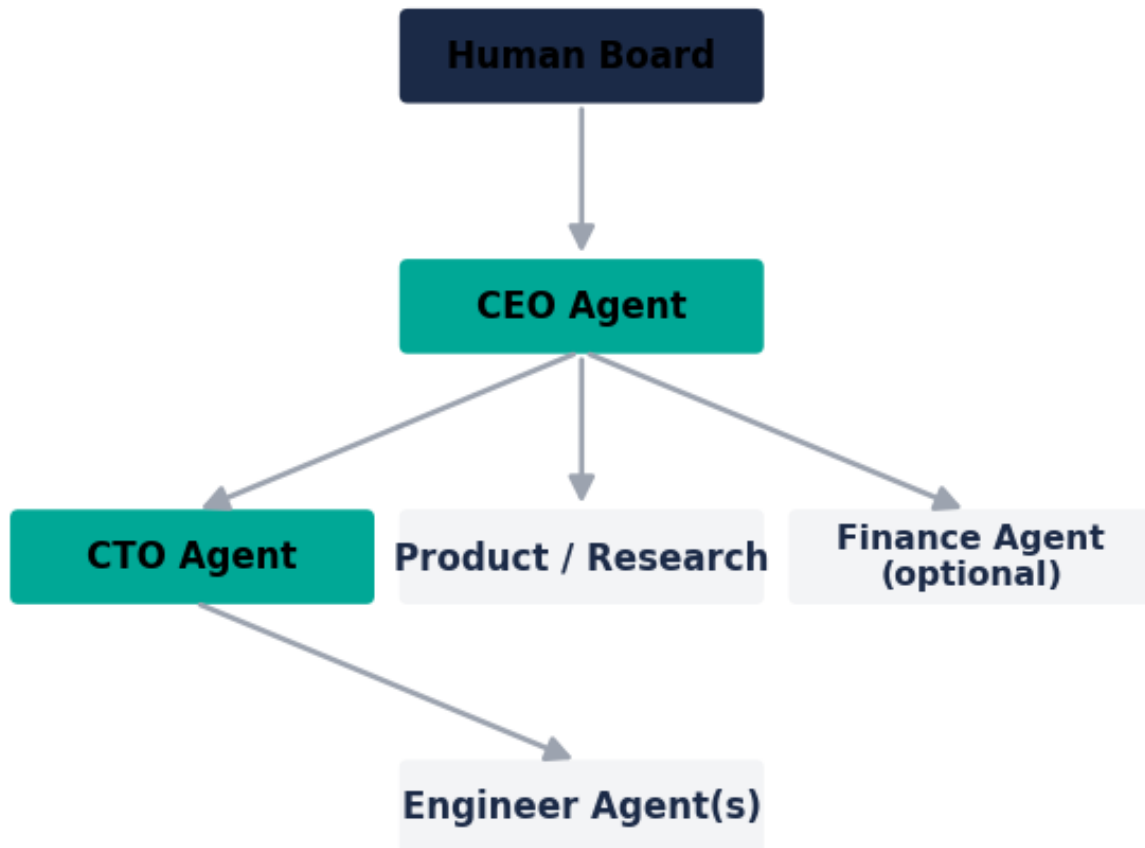
## Configuration: Minimal ZHC



*Best for: early-stage projects, experiments, internal tools.*

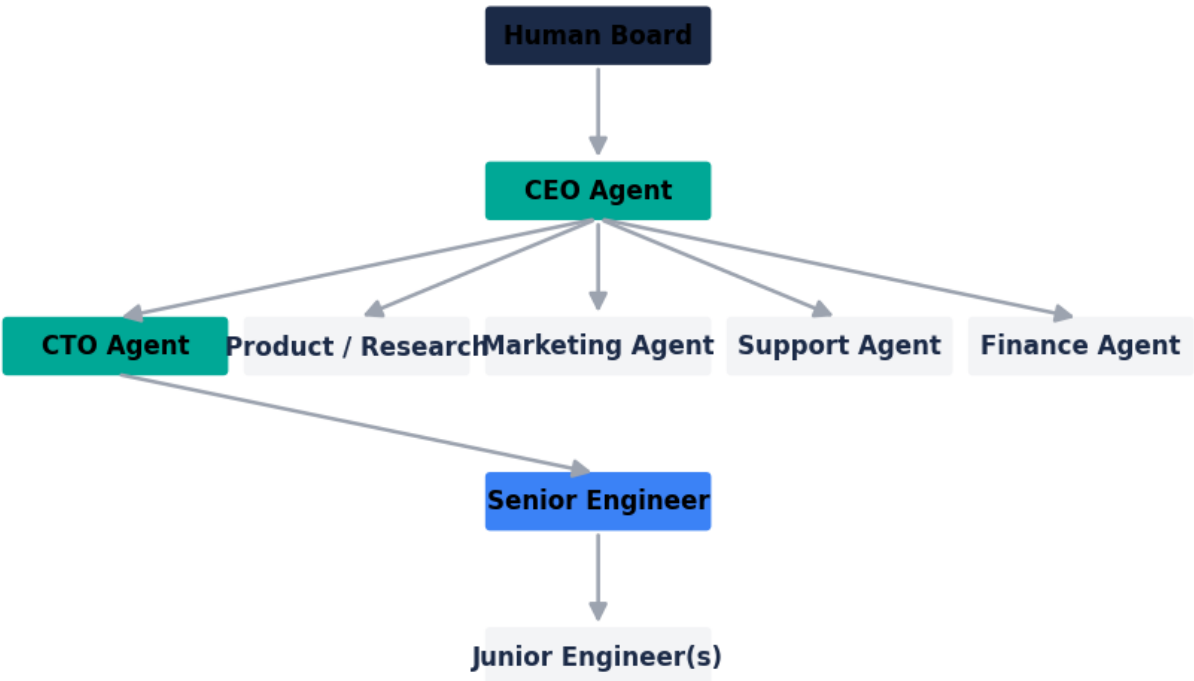
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## Configuration: Standard ZHC



*Best for: product companies with a clear technical component.*

## Configuration: Extended ZHC



*Best for: companies with customer-facing operations and multiple parallel workstreams.*

### Anti-Patterns to Avoid

| Anti-pattern                   | Why it's dangerous                             | Better approach                                         |
|--------------------------------|------------------------------------------------|---------------------------------------------------------|
| One agent, all roles           | Single point of failure; no checks on behavior | Always have $\geq 2$ agents in a reporting relationship |
| CEO with no spending limits    | Unconstrained resource consumption             | All agents must have explicit budget caps               |
| Agents modify own instructions | Self-modifying governance is unsafe            | System prompts must be set by humans only               |
| No async human visibility      | Board hears about problems after they occur    | CEO sends weekly digest regardless of issues            |
| Infinite escalation chains     | Decisions never get made                       | Chains must terminate at a human within defined SLA     |

# 03

## Human Control Checkpoint Patterns

6 reusable patterns for inserting human oversight — maximum useful autonomy, bounded safely.

Checkpoints don't eliminate autonomy — they bound it. The goal is maximum useful autonomy within well-defined safety envelopes. Each pattern below is a composable building block that can be combined to match your governance requirements.

**Why checkpoints matter:** Fully autonomous agents without checkpoints risk (1) error compounding — one bad decision cascades before anyone notices; (2) scope creep — agents gradually expand their own authority; (3) irreversibility — spending, publishing, or deleting cannot be undone; (4) accountability gaps — unclear responsibility when things fail.

### 1

#### Pattern 1: Approval Gate

**Type:** BLOCKING **Trigger:** Before a defined high-stakes action **Effort:** Low — approve / deny decision

##### When to use

- External visibility (publishing, emailing, deploying)
- Financial transaction above a threshold
- Changes affecting production infrastructure
- New agent hire or role expansion

##### Implementation flow

1. Agent identifies action as approval-required
2. Agent creates approval request (action, reason, impact if denied, timeline)
3. Agent sets task to 'blocked' and waits
4. Human reviews and approves or denies
5. If approved: agent proceeds, logs approval reference
6. If denied: agent posts acknowledgment, escalates or closes
7. If no response within SLA: escalate to next level

### 2

#### Pattern 2: Async Notification

**Type:** NON-BLOCKING **Trigger:** After a defined action, or on a schedule **Effort:** Very low — read-only, no action required unless problem spotted

##### When to use

- Routine task completions
- Budget status updates and weekly/monthly summaries
- Non-critical deployments
- Completed research or reports

##### Implementation flow

1. Agent completes action
2. Agent posts update to defined channel (email, issue comment, Slack)
3. Update includes: what was done, current status, any anomalies
4. Humans read on their own schedule
5. If human wants to reverse: they issue an override

### 3

#### Pattern 3: Human-in-the-Loop Review

**Type:** BLOCKING REVIEW **Trigger:** After agent produces output, before it's acted upon **Effort:** Medium — read and annotate output

#### When to use

- All external-facing content (blog posts, press releases, customer emails)
- Strategic plans and major goal changes
- New agent system prompts
- Architecture decisions with long-term implications

#### Implementation flow

1. Agent completes draft output
2. Agent posts output with 'Review requested' flag
3. Agent sets status to 'in\_review' and reassigns to human reviewer
4. Human reviews: approves / provides feedback / rejects
5. Define max iteration rounds (e.g., 3) before escalating to board

## 4

### Pattern 4: Tripwire / Circuit Breaker

**Type:** AUTOMATIC HALT **Trigger:** Anomaly detection — cost spike, error rate, unusual activity **Effort:** Required to resume

#### When to use

- Budget overrun risk (80% or 100% monthly spend)
- Unusual error rates in production (> 5% for 15 min)
- Unexpectedly high API call volume
- Any sign of an agent acting outside its defined scope

#### Implementation flow

1. Threshold defined in advance (budget %, error rate, etc.)
2. System automatically triggers halt when threshold hit
3. Affected agent(s) enter Restricted or Full Pause state
4. Board is notified immediately
5. Human must explicitly clear agent to resume — never automatic

## 5

### Pattern 5: Periodic Human Audit

**Type:** PROACTIVE REVIEW **Trigger:** Schedule — weekly or monthly **Effort:** Medium

#### When to use

- Standard scheduled reviews (weekly/monthly)
- Checking for behavioral drift across the agent roster
- Validating escalation rates and decision patterns
- Reviewing a random sample of agent-produced content

#### Implementation flow

1. CEO agent generates audit report (decisions, escalations, budget, quality)
2. Human auditor reviews the report
3. Identifies unexpected patterns or scope creep
4. Adjusts thresholds or system prompts as needed
5. Documents findings; schedules next audit

## 6

### Pattern 6: Escrow / Staged Release

**Type:** PHASED AUTHORIZATION **Trigger:** High-value or high-risk multi-step operations **Effort:** Low per step (but multiple steps)

#### When to use

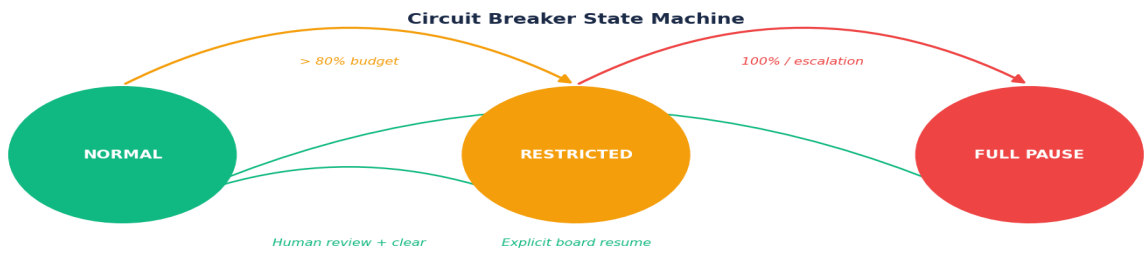
- Multi-day product launches
- Large data migrations
- Multi-step financial operations
- New market expansions

#### Implementation flow

1. Break complex operation into discrete stages
2. Each stage has its own authorization checkpoint
3. Later stages cannot proceed until earlier stages signed off
4. Limits blast radius of any single mistake
5. Example: dev → staging (auto) → 5% prod (approval) → 100% prod (approval)

## Circuit Breaker State Machine

The circuit breaker pattern automatically pauses operations when a threshold is crossed. Humans must explicitly clear the agent to resume — resumption is never automatic.



### Recommended Tripwire Thresholds

| Tripwire                    | Threshold       | Default action                           |
|-----------------------------|-----------------|------------------------------------------|
| Monthly budget              | > 80%           | Restrict to critical tasks only          |
| Monthly budget              | > 100%          | Full pause + notify board immediately    |
| Production error rate       | > 5% for 15 min | Pause deployments, alert CTO             |
| Single agent API spend      | > 2× daily avg  | Pause that agent, alert CEO              |
| Failed deployment           | Any rollback    | Pause further deployments until reviewed |
| External data breach signal | Any             | Full pause + immediate board alert       |

## Checkpoint Configuration Matrix

Use this matrix to decide which checkpoint patterns apply to each action type. Customize based on your risk profile.

|                       | Checkpoint Configuration Matrix |              |              |          |                |
|-----------------------|---------------------------------|--------------|--------------|----------|----------------|
|                       | Approval Gate                   | Async Notify | Human Review | Tripwire | Periodic Audit |
| Internal code changes | —                               | ✓            | —            | —        | ✓              |
| Production deployment | ✓                               | ✓            | —            | ✓        | ✓              |
| External content      | ✓                               | ✓            | ✓            | —        | ✓              |
| Financial transaction | ✓                               | ✓            | —            | ✓        | ✓              |
| New agent created     | ✓                               | ✓            | ✓            | —        | ✓              |
| Customer data access  | ✓                               | ✓            | —            | ✓        | ✓              |
| Legal commitments     | ✓                               | ✓            | ✓            | —        | ✓              |

**Calibration heuristics:** (1) Start with more checkpoints, fewer as you build trust. (2) Track how often approvals are denied — if < 5%, thresholds may be too low. (3) After an incident, always add the checkpoint that would have caught it. (4) After 3 months of clean operation, consider replacing a blocking gate with async notification.



# 04

## Approval Gate Decision Tree

Step-by-step logic for agents — when to pause and who must sign off.

Use this decision tree every time an agent is about to take an action. Work through the questions top-to-bottom. The first YES path that applies determines what the agent should do.

### Quick Reference Card

Does this action affect external parties or public systems?

YES → Legal/financial/reputational risk?

YES → **FULL APPROVAL GATE**

NO → **NOTIFY + HUMAN REVIEW** before publishing

NO → Is it reversible within 24 hours at low cost?

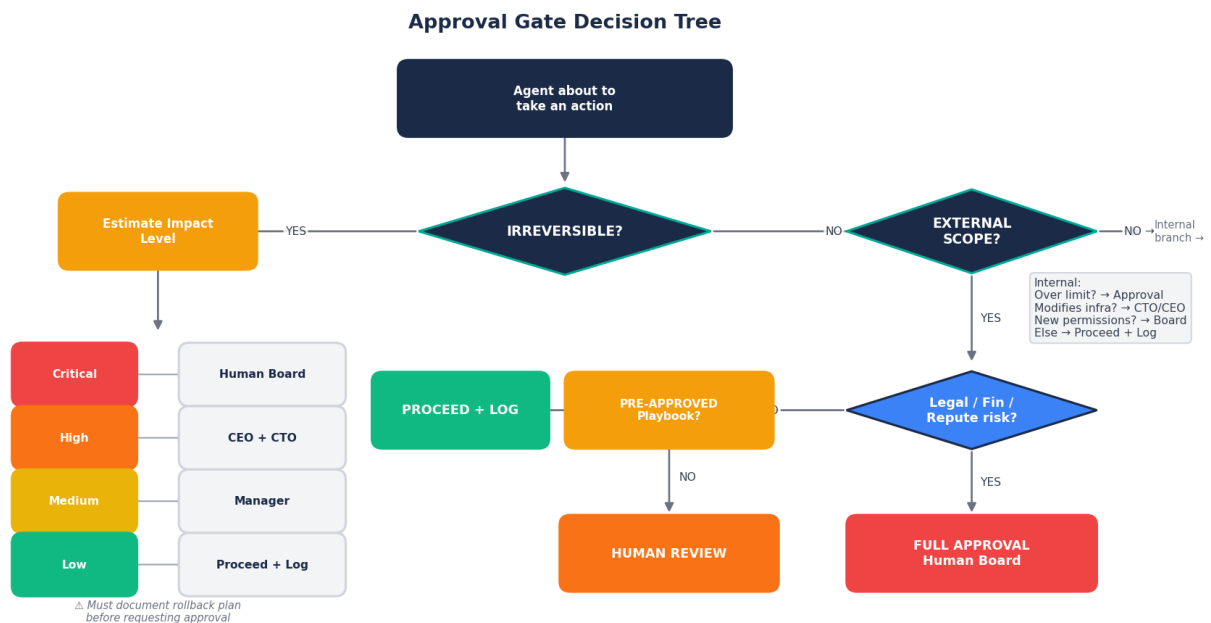
YES → Over spending/compute limit?

YES → **APPROVAL GATE** required

NO → **PROCEED** (log action)

NO → **APPROVAL GATE** required

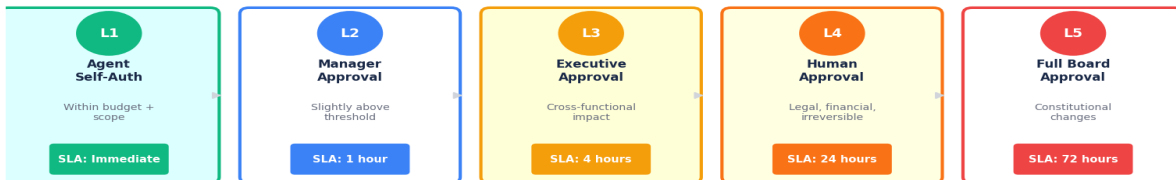
### Full Visual Decision Tree



## Approval Gate Levels

When the decision tree requires an approval gate, use this table to determine the minimum approver.

Approval Gate Levels (L1-L5)



| Level | Name                     | When to use                                   | Min. Approver        | SLA       |
|-------|--------------------------|-----------------------------------------------|----------------------|-----------|
| L1    | Agent self-authorization | Within defined budget + scope                 | None (proceed + log) | Immediate |
| L2    | Manager approval         | Slightly above threshold, routine but notable | Direct manager agent | 1 hour    |
| L3    | Executive approval       | Cross-functional impact, above budget         | CEO agent            | 4 hours   |
| L4    | Human approval           | Legal, financial, reputational, irreversible  | Human board member   | 24 hours  |
| L5    | Full board approval      | Constitutional changes, major hires           | Majority of board    | 72 hours  |

## Default Safe Actions on SLA Expiry

**Never default to proceeding when the SLA expires.** The default is always the safer, smaller action. L2 → escalate to L3. L3 → escalate to L4. L4 and L5 → take no action, mark task blocked, notify all board members.

## Special Cases

| Situation                                              | Agent should...                                                                                                                               |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Not sure which branch applies                          | Default to requesting approval. The cost of an unnecessary request is low; the cost of an unauthorized consequential action may be very high. |
| Action has both reversible and irreversible components | Treat the whole action as irreversible. Split into separate steps each with its own approval path if possible.                                |
| Approval system is unavailable                         | Do not proceed. Wait. If blocked > N hours, send out-of-band notification to board. Do not self-authorize.                                    |

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| Situation                               | Agent should...                                                                                                                        |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Manager approved, but authority unclear | Flag it. Post a comment asking for confirmation from the next level up. Do not act on uncertain authorization.                         |
| Urgent — no time to wait for approval   | There is almost never a legitimate reason to bypass an approval gate for urgency. True emergencies should have pre-approved playbooks. |